

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

ANTENNAS & MICROWAVE ENGINEERING

(Electronics and Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Define Omni directional antenna. | 2M |
| (b) List out the far field equations, E_θ and H_ϕ of a short dipole. | 2M |
| (c) Mention the applications of helical antennas. | 2M |
| (d) State Fermat's principle. | 2M |
| (e) Mention the expression for spacing between the elements in EFA with increased directivity. | 2M |
| (f) Define dominant and degenerate modes in waveguide. | 2M |
| (g) What is the coupling factor of a directional coupler? | 2M |
| (h) What is a magic tee? | 2M |
| (i) Mention the applications of IMPATT diode. | 2M |
| (j) Draw the block diagram of microwave bench setup. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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| 2 (a) Explain the working principle of Yagi – Uda antenna with suitable sketches. | 5M |
| (b) Derive the relation between directivity and effective aperture area of an antenna. | 5M |
| OR | |
| 3 (a) Explain the radiation mechanism of a two-wire antenna. | 5M |
| (b) Given $\bar{W}_{rad} = A_o \sin^2 \theta \frac{\bar{a}_r}{r^2}$, W/m^2 . Find the directivity. | 5M |
| 4 (a) Explain the geometry and radiation Mechanism of a Rectangular patch antenna. | 5M |
| (b) With a neat sketch explain the geometry of non-metallic dielectric lenses. | 5M |
| OR | |
| 5 (a) List out the advantages and limitations of Micro strip antennas. | 5M |
| (b) Explain the radiation mechanism of a Parabola reflector. | 5M |
| 6 (a) Derive the field equations of rectangular waveguide in TE mode. | 7M |
| (b) Derive the beam width of a End fire array. | 3M |
| OR | |
| 7 (a) Derive the array factor of two element uniform linear array. | 5M |
| (b) Derive the expression for cutoff frequency, phase constant and phase velocity of a wave in a circular wave guide. | 5M |
| 8 (a) Explain the operating principle of isolator and discuss its working with relevant diagram. | 5M |
| (b) Draw and explain the operation of a Reflex Klystron oscillator. | 5M |
| OR | |
| 9 (a) Explain about magnetron oscillator. | 5M |
| (b) Write a short note on two hole couplers. | 5M |
| 10 (a) Draw and explain the working of TRAPATT diodes. | 5M |
| (b) Explain the method of measurement of high VSWR. | 5M |
| OR | |
| 11 (a) Describe briefly the equipment that is used to measure impedance using a slotted line. | 5M |
| (b) Write the principle and operation of GUNN oscillator. | 5M |
